

DiaTool-DPO: Enhancing Tool-Augmented LLMs via Direct Preference Optimization

MDP-grounded trajectory preferences for reliable multi-turn tool

Audience: ML researchers and NLP engineers in TA-LLMs & RLHF alignment

Technical presentation (8 slides, 16:9)

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Core Problem

SFT-only TA-LLMs struggle with:

- Incomplete queries (missing arguments)
- Out-of-scope user requests
- Multi-turn dialogue control errors

Need: trajectory-level preference optimization

to teach when to ask, call, or reject tools.

Failure 1

Fails to ask follow-up slot questions

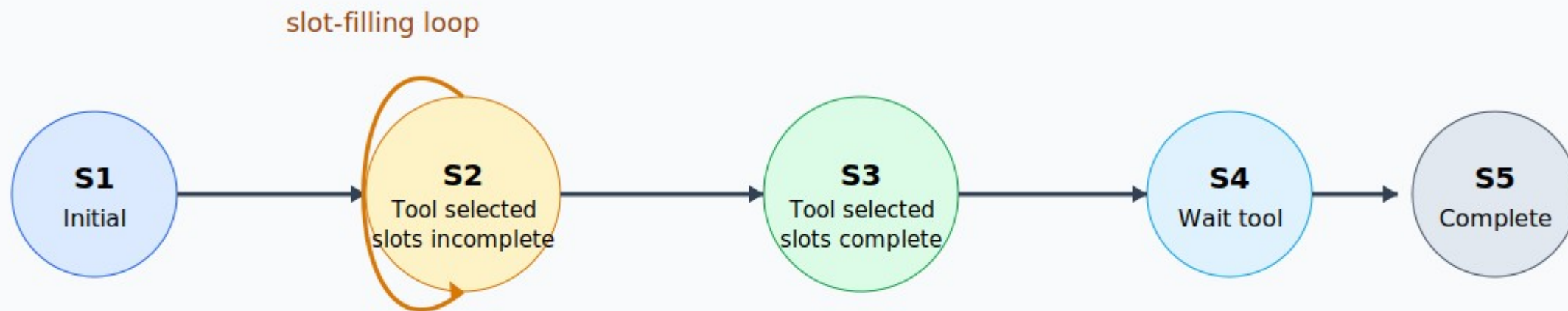
Failure 2

Hallucinates or prematurely invokes tools

Failure 3

Fails to reject inappropriate tool calls

Modeling TA-LLM Dialogue as a 5-State Markov Decision



Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Brief Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	“What is the weather in Boston today?” All required slots provided
Type 2	$1 \rightarrow (2 * N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	“Book me a flight” Requires iterative slot clarification
Type 3	$1 \rightarrow 1$	“Write a poem about rainbows” Out-of-tool scope, reject tool call

Each type induces unique preference signals for chosen vs. rejected trajectories.

Automatic Paired Trajectory Construction Without Human

Construction Pipeline

1) Collect dialogues by query type

2) Build chosen/rejected pairs

3) Encode lessons:

- Prevent redundant slot-filling
- Prevent slot hallucination
- Learn proper tool reject/accept

Type	Chosen	Rejected	Total / Diff.
Type 1	5,283	4,435	9,718 easy
Type 2	4,524	4,742	9,266 medium
Type 3	6,194	5,930	12,124 hard

Total preference pairs: 31,108

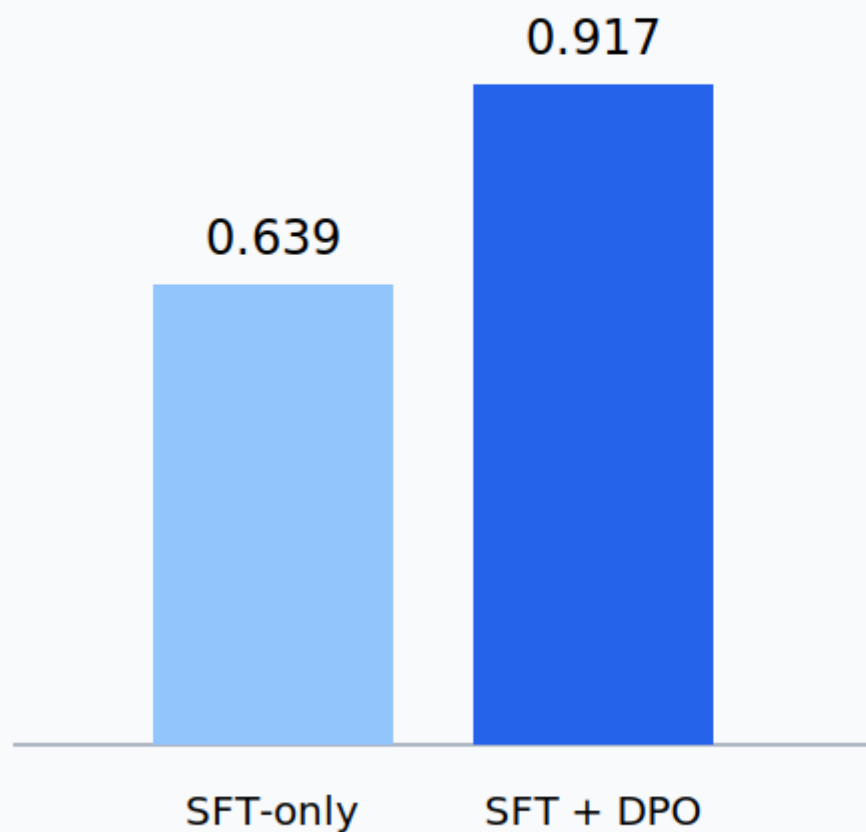
SFT + DiaTool-DPO Achieves Best Macro Average Across All

Model	Training	Call	Completion	Slot	Relevance	Micro	Macro
Prop.-8B	DiaTool-DPO-only	0.243	0.554	0.194	0.348	0.335	0.335
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

SFT + DiaTool-DPO achieves the best Macro Avg across all model families.

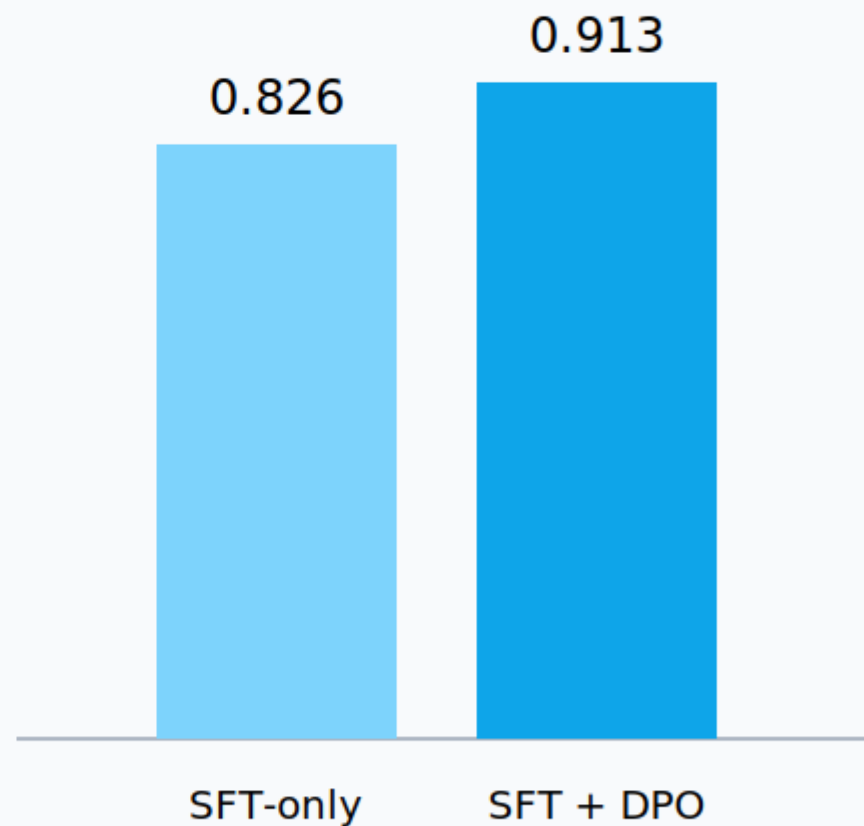
LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relev

Slot Accuracy



+43.5%

Relevance



+10.5%

Key Takeaways: RL-Based Dialogue Control for Pro

- 1) 5-state MDP enables systematic trajectory preference construction.
- 2) Automatic chosen/rejected pairing removes manual preference labeling.
- 3) DPO complements (not replaces) SFT for dialogue control.
- 4) Largest gains appear in slot-filling and relevance/rejection behavior.
- 5) Language-agnostic method supports high-quality smaller-model deployment.